Extending SensorQA with Location-Aware and Context-Aware Computing

Project extension plan for Challenge 1: Grounded Activity Question-Answering

1. Where the project stands

Three facts from the Challenge 1 report shape what to add:

- **GPS is currently an out-of-scope rejection.** The safety layer rejects food, heart rate and GPS queries. Adding location means moving GPS from "reject" to "answer", while food and heart rate stay rejected.
- **Perception is the weak point.** Weighted F1 is 0.6537, Standing in place is 0.33, and Running and Bicycling have precision around 0.22 to 0.27. Standing & moving has 0 test support in Fold 0. Duration MAE is 598.4 s.
- **The architecture is easy to extend.** The QueryPlan JSON, the deterministic engine and the 7-field schema are modular, so new capabilities can be added as new intents.

2. What can be added

Requested features

Feature	What it means in SensorQA	Example questions
Context-aware	Infer the situation around the activity (at home, at work, in a car, with friends, indoors) and use it to disambiguate activities	"Was I walking while at work?", "How long was I sitting in a car?"
Location-aware	Place semantics (home, work, gym, outdoors) and mobility (stationary vs. moving, travel mode, distance)	"How long was I at the gym?", "Did I walk to the place where I ate?"

Other additions that fit the architecture

- **Context-conditioned classification:** use context and location as priors to fix confusions such as Sitting vs. Lying, or Bicycling vs. in-vehicle motion.
- **Transportation mode detection:** walk, bike, bus, car, train; a natural bridge between location and motion.
- **Multi-turn QA:** "What about the afternoon?" using dialogue state.

- **Routine and anomaly questions:** "Is this a typical day for me?" using deterministic baselines.
- **Personalization:** per-user fine-tuning or calibration, which helps cross-subject generalization.
- **Better imbalance handling:** focal loss, resampling and per-class thresholds for Running and Bicycling.
- **Uncertainty-aware answers:** surface confidence in the answer ("about 700 s, low confidence").

3. Free and open-source dataset options

Licenses and access terms vary, so check each one before use.

Dataset	Gives you	Use in the project
ExtraSensory (already used)	Besides motion, it includes context labels (e.g., at home, at work, in a car, with friends, indoors/outdoors) and location-derived features	**Start here.** Enables context-aware work with no new data. Check which location fields are public, since it may release derived summaries rather than raw coordinates.
	Multi-sensor phone data (including GPS and Wi-Fi) with transportation modes	Travel-mode and location-aware reasoning (academic registration may be required)
GeoLife (Microsoft Research)	GPS trajectories, many with transport-mode labels	Trajectory logic, stay-point detection, distance and speed
StudentLife (Dartmouth)	Phone sensing including GPS, activity, conversation and sleep over weeks	Longer-horizon context and routines
OpenStreetMap (Overpass/Nominatim)	Free map and POI data	Convert coordinates to place types (restaurant, gym, park, road)
Foursquare Open Source Places	Open POI data	Alternative or complement to OSM

Dataset	Gives you	Use in the project
CASAS smart-home datasets	Ambient sensors with room-level location and activities	Indoor location awareness (which room)
UCI HAR, WISDM, MotionSense, PAMAP2, HHAR	Standard activity datasets	Pretraining, extra Running/Bicycling samples, cross-dataset tests
UJIIndoorLoc	Wi-Fi fingerprints for indoor positioning	Indoor localization experiments
SensorQA benchmark (built on ExtraSensory)	Human-written QA pairs over daily-life sensing	External benchmark for QA quality (verify current availability)

Practical suggestion: use ExtraSensory as the primary dataset, since it already has motion and context together. Use OSM or Foursquare for place semantics and GeoLife/SHL for mobility logic. Avoid mixing many datasets early, because label spaces and sensor setups differ.

4. Project process

Phase 0: Audit and scoping (week 1)

- Inventory which context labels and location fields ExtraSensory provides, with per-label coverage and class balance.
- Decide the final scope of new question types and the new rejection list.

Phase 1: Data layer (weeks 2–3)

- Align context labels and location features to the 20 s windows and the Activity Timeline Store.
- Build a Context Timeline and a Location Timeline alongside the Activity Timeline.
- Handle missing modalities explicitly, since many windows will lack location.

Phase 2: Perception upgrades (weeks 3–5)

- Add a multi-task context head to Model A (multi-label, since contexts overlap).
- Add location features: stationary vs. moving, speed or movement-diameter proxies, and place type via OSM/POI when coordinates exist.
- Fuse these with the 98 kinematic features and retrain with temperature scaling.

- Re-evaluate per-class F1, especially the weak classes.

Phase 3: Reasoning layer (weeks 5–7)

- Extend the QueryPlan schema with new intents: `context_filter`, `location_duration`, `place_transition`, `co_occurrence` (activity AND context).
- Add deterministic operators: interval intersection (activity ∩ context), dwell time, place-visit counts.
- Update the safety policy: GPS and context become answerable, while food and heart rate stay rejected.

Phase 4: Answer schema (week 7)

- Extend the 7-field output with **Context** and **Location** fields, or keep 7 fields and put them in Evidence/Explanation if the challenge requires the fixed schema.

Phase 5: Evaluation (weeks 8–9)

- Task-level metrics for the new question types: accuracy, duration MAE, interval IoU.
- Ablations: motion only vs. motion + context vs. motion + context + location.
- Cross-subject testing across all 5 folds (not just Fold 0), with variance.
- Rejection tests for out-of-scope and unanswerable-from-data queries.

Phase 6: Edge and privacy (week 10)

- Re-profile model size and latency with the added heads.
- Add privacy handling: coarse location, on-device processing, no raw coordinates sent to the LLM.

Phase 7: Documentation (week 11)

- Update the report with new architecture diagrams, math and results.

5. Key risks

- **Label sparsity:** many ExtraSensory context labels are weakly or sparsely annotated, so expect noisy supervision and evaluate carefully.
- **Location availability:** if raw coordinates aren't public, place-level reasoning will rely on derived features, which limits what can be claimed.
- **Honest reporting:** since Fold 0 is one split, report all folds before claiming improvements.